AidLedger: A Decentralized GRC Framework for Transparent Aid Distribution

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Project Abstract

Government aid programs, such as the Benazir Income Support Program (BISP) and Ehsaas in Pakistan, provide financial assistance to low-income households. However, these initiatives frequently face issues of corruption, ineligible beneficiaries, and limited transparency in the distribution process. To address these challenges in a more general and adaptable way, this project proposes the development of a decentralized Governance, Risk, and Compliance (GRC) framework inspired by State Bank of Pakistan (SBP) regulations. The framework is dejsigned to ensure that only genuinely deserving beneficiaries are registered, verified, and provided with financial aid in a transparent and accountable manner.

The proposed system leverages blockchain technology to immutably record and enforce eligibility policies. A dummy bank, simulating real-world institutions such as the National Bank of Pakistan or HBL, will interact with the blockchain and comply with all rules encoded in the framework. During registration, a beneficiary will provide data such as CNIC, income, and household size, which will be verified through simulated NADRA and other related entities. If the applicant meets the required criteria, a prototype debit card (physical or virtual) will be issued, directly linked with blockchain-enforced rules. This card will enable controlled transactions such as limited cash withdrawals and grocery purchases from authorized vendors.

Additionally, the framework supports dynamic policy updates. For example, if income eligibility thresholds are revised, the system will automatically re-evaluate all registered beneficiaries. Those who no longer qualify will be flagged, and automated notifications will be delivered in Urdu, ensuring transparency, accountability, and compliance.

1. Introduction

The Government of Pakistan operates large-scale aid programs like BISP and Ehsaas to alleviate poverty. These programs distribute billions of rupees annually, yet inefficiencies and corruption hinder their impact. At present, banks verify beneficiaries using Know Your Customer (KYC) procedures under SBP guidelines, but records are centralized, making them vulnerable to tampering.

This project introduces a blockchain-based decentralized GRC framework that replicates the regulatory model of SBP and banks. The system will encode rules such as income thresholds, household size, and verification checks into smart contracts, ensuring transparency, auditability, and immutability. Banks (simulated as dummy institutions) will interact with the blockchain to register and verify beneficiaries.

Key features include:

1. Immutable policy enforcement using blockchain smart contracts.
2. Dynamic updates so new policies automatically propagate to all beneficiaries.
3. Beneficiary registration and KYC verification through simulated NADRA checks.
4. Prototype debit card issuance connected to blockchain rules, usable for restricted cash withdrawal and grocery purchases.
5. Automated Urdu notifications to beneficiaries upon policy updates.

Sub-tasks include:

1. Designing the GRC framework.
2. Preparing SRS and SDS documents to formalize requirements and design.
3. Developing smart contracts for eligibility verification and policy management.
4. Building a dummy bank interface to simulate aid distribution.
5. Designing the card issuance process and reader integration (virtual or hardware).
6. Simulating external entities like NADRA and SBP.
7. Developing a notification module for policy changes.

At the end, success will be tested by demonstrating registration, eligibility verification, card issuance, and policy updates on a simulated setup with blockchain enforcement.

2. Success Criterion

The project will be considered successful if:

1. A GRC framework is designed based on SBP regulations.
2. Blockchain smart contracts are implemented to enforce eligibility rules immutably.
3. A dummy bank system interacts with blockchain to verify and approve beneficiaries.
4. Beneficiaries can be registered and dynamically validated against updated policies.
5. A prototype card (virtual or NFC-based) is issued and integrated with the framework.
6. Urdu notifications are generated for beneficiaries upon eligibility status changes.

3. Related Work

Blockchain applications in governance and aid distribution have been studied globally. Historically, aid programs relied on centralized record-keeping, which made them prone to inefficiency and corruption. The World Food Programme (WFP) piloted blockchain-based food distribution in Jordan refugee camps (Building Blocks Project), reducing fraud and costs [1]. Research also shows that blockchain ensures transparency in social welfare programs by providing immutable and auditable ledgers [2].

In Pakistan, studies highlight weaknesses in BISP's distribution process, citing ghost beneficiaries and mismanagement [3]. Internationally, financial institutions increasingly adopt blockchain for KYC compliance and GRC frameworks [4], aligning with our approach.

However, most prior work is conceptual or industrial-scale; few academic projects simulate both governance frameworks and card-based disbursements. This project fills that gap by combining a GRC model + blockchain policies + card prototype for a university-level demonstration.

4. Project Rationale

Inefficiencies and lack of transparency in government aid distribution programs often result in mismanagement and exclusion of deserving beneficiaries. This project aims to develop a decentralized blockchain-based GRC framework to address these challenges, demonstrating secure, transparent, and auditable aid disbursement processes.

Issue	Current BISP	Our System	Key Point
Corruption	Rs. 141billion+ lost [10]	Smart contracts block human theft	No humans = No theft
Fake Users	Ghost accounts steal money	Blockchain stops fake entries	One person = One account only
Policy Changes	Takes months to update	Updates in seconds automatically	Instant changes for all users
Money Safety	Unauthorized deductions happen	Smart contracts protect all funds	Money goes where it should
Transparency	Hard to track money flow	All transactions recorded permanently	Complete audit trail always

4.1 Aims and Objectives

- To design a blockchain-based GRC framework ensuring fair and transparent aid distribution.
- To simulate SBP regulations and bank compliance in a decentralized architecture.
- To implement dynamic eligibility rules using smart contracts.
- To demonstrate beneficiary registration, verification, and card issuance.
- To explore card prototype approaches (virtual, NFC, smart card) and select the most feasible.

4.2 Scope of the Project

- **In-scope:**
  - ✓ Blockchain smart contracts for eligibility enforcement.
  - ✓ Dummy bank integration.
  - ✓ Simulation of NADRA/SBP data.
  - ✓ Prototype card issuance (virtual or hardware).
  - ✓ Urdu notifications on policy updates.

- **Out-of-scope:**
    - Real NADRA or SBP integration (simulated with dummy data).
    - Industrial-grade debit card manufacturing.
    - Advanced EMV secure chip implementations (beyond FYP scope).

5. Proposed Methodology and Architecture

This section provides insight into the methodology and architecture employed in the development of the envisioned decentralized aid distribution system. The approach is systematic and structured to demonstrate immutable rule enforcement, eligibility verification, and dynamic policy updates. The methodology combines theoretical design with practical demonstration, and includes step-by-step procedures, flowcharts, and architectural modules.

Methodology

The project will follow a two-phase approach to ensure a logical sequence of development and demonstration:

Phase 1 (Sep–Dec 2025): System Foundations and Blockchain Implementation

1. Conduct literature review and requirement analysis for government aid programs.
2. Study SBP regulations to understand rules, compliance requirements, and eligibility criteria.
3. Design a GRC framework to encode SBP policies for eligibility and transaction rules.
4. Prepare Software Requirements Specification (SRS) and Software Design Specification (SDS) covering functional and non-functional requirements.
5. Create dummy datasets simulating NADRA and bank records for testing eligibility rules.
6. Develop smart contracts on blockchain to store policies and enforce eligibility rules immutably.

Outcome: Phase 1 ensures the core technical contribution of blockchain-based eligibility validation with dummy datasets, demonstrating immutability and policy enforcement.

Phase 2 (Jan–May 2026): Card Prototype, POS Demo, and System Integration

1. Build a dummy bank interface to simulate registration, verification, and card issuance workflows.
2. Design a card issuance system linked to blockchain rules for real-time eligibility enforcement.
3. Explore three card approaches (feasible option selected for prototype):
   - Virtual card (mobile/web-based ID with optional PIN).
   - NFC card + reader (e.g., ACR122U integrated with blockchain).
   - Smart card with secure element (only if feasible).
4. Implement a POS demo, demonstrating that card/ID transactions are validated according to blockchain-enforced rules.
5. Add Urdu notification module to inform beneficiaries of eligibility changes, policy updates, and transaction results.
6. Perform final integration and testing, ensuring all system components work cohesively: registration portal, database, blockchain, dummy bank, card, POS, and notifications.

Outcome: Phase 2 delivers a fully integrated system demonstrating end-to-end eligibility verification, card prototype usage, and dynamic policy enforcement.

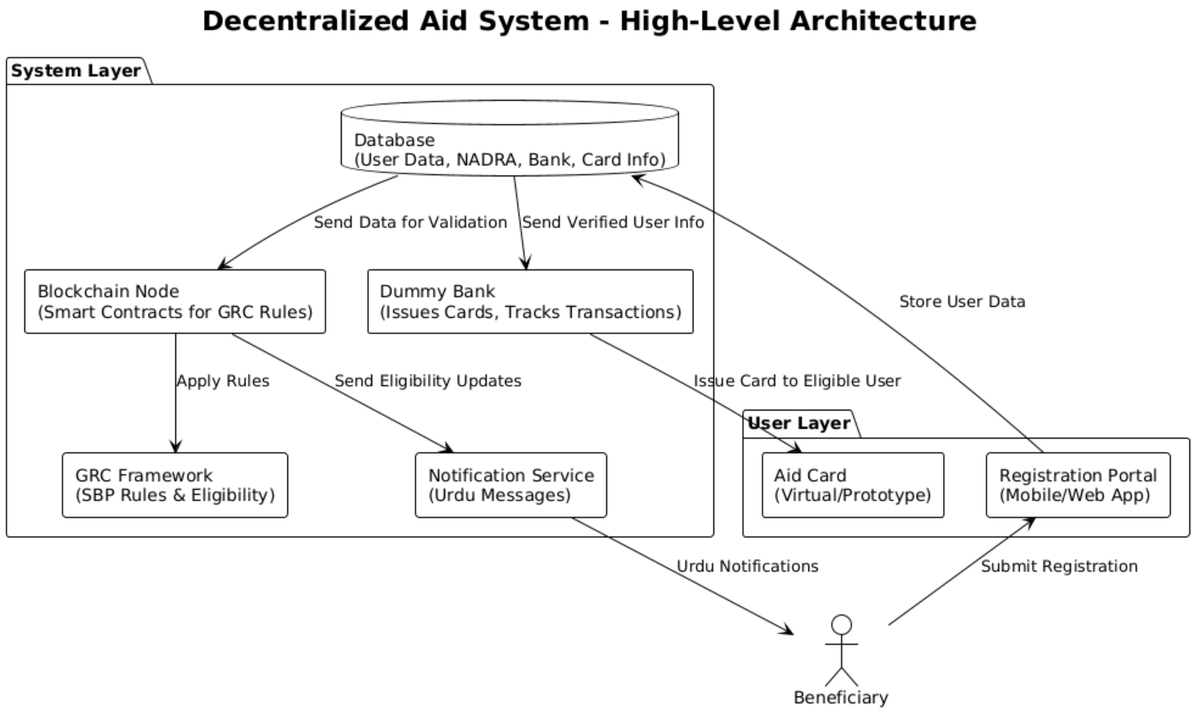
System Architecture Overview

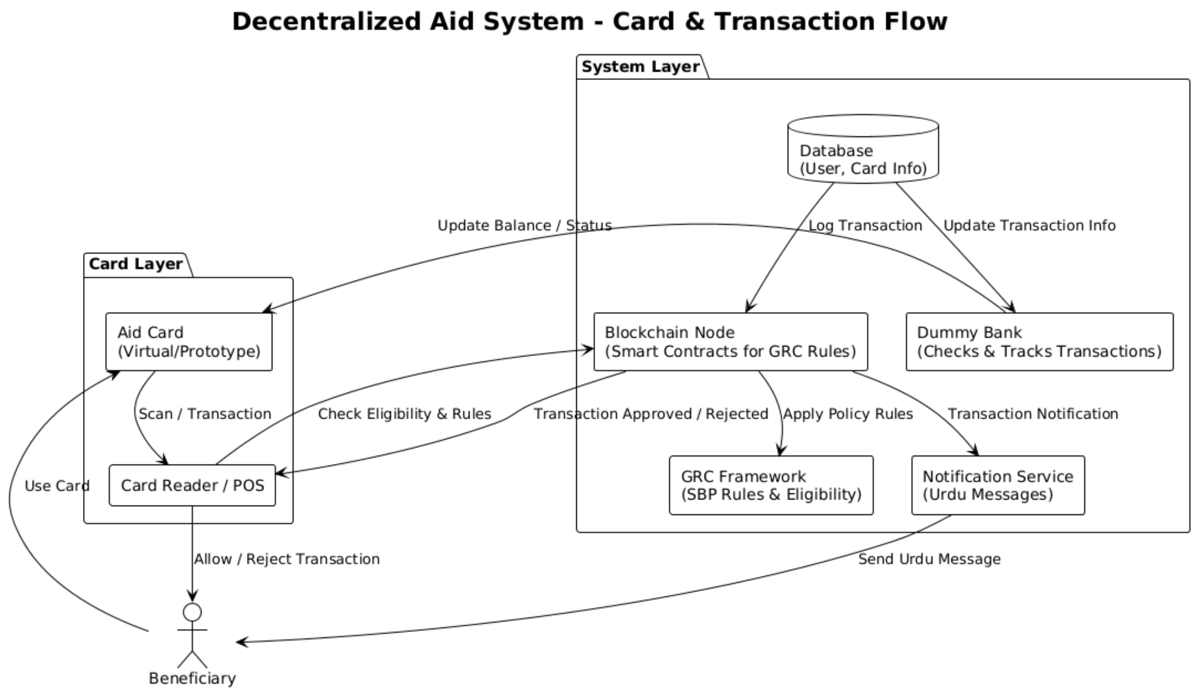
The architecture of the system is organized into three main layers:

1. User Layer:
    - Beneficiaries interact via Registration Portal and use the issued aid card for transactions.
2. System Layer:
    - Database: Stores registration data, dummy NADRA records, bank data, card information, and transaction logs.
    - Blockchain Node: Executes smart contracts enforcing GRC rules immutably.
    - GRC Framework Module: Encodes SBP policies for eligibility and transaction compliance.
    - Dummy Bank Module: Verifies users, issues cards, and tracks transactions.
    - Notification Service: Sends notifications in Urdu for eligibility, policy updates, and transaction confirmations.
3. Card & Transaction Layer:
    - Aid Card and Card Reader/POS: Validate transactions against blockchain-enforced rules.
    - POS Demo: Demonstrates real-time enforcement of eligibility, spending limits, and allowed merchant access.

This methodology ensures dynamic, secure, and transparent enforcement of rules, where policy updates automatically propagate through blockchain validation, card usage, and notifications. By completing blockchain-based validation in Phase 1, the system demonstrates the core technical contribution, while Phase 2 focuses on the card prototype and full integration.

System architecture diagram:



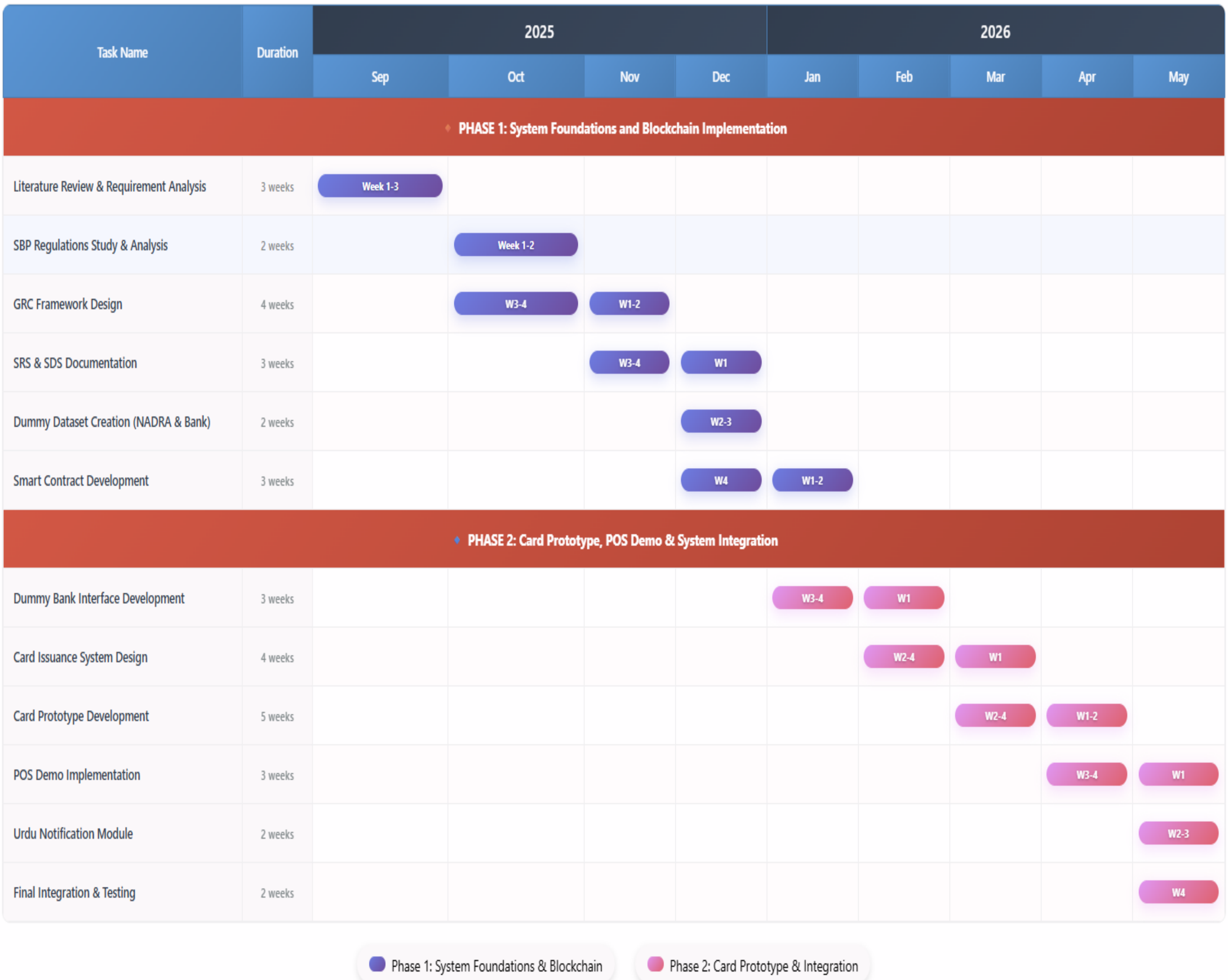


6. Individual Tasks (Tentative)

Team Member	Activity	Tentative Date
Muhammad Ali	Conduct literature review & requirement analysis	Sep–Oct 2025
	Study SBP regulations & design GRC framework	Oct 2025
	Prepare SRS document	Oct–Nov 2025
	Create dummy datasets simulating NADRA & bank records	Nov–Dec 2025
	Assist smart contract development & blockchain validation	Dec 2025–Jan 2026
	Support card prototype design & POS demo integration	Jan–Mar 2026
	Perform final integration & testing	Apr–May 2026
	Prepare final demo & presentation materials	May 2026
Usman Ghani	Conduct literature review & requirement analysis	Sep–Oct 2025
	Prepare SDS document (architecture, database design, flowcharts)	Nov–Dec 2025
	Develop smart contracts for policy storage & eligibility verification	Dec 2025
	Build dummy bank interface for registration & verification	Jan–Feb 2025
	Assist card prototype design & POS demo integration	Jan–Mar 2026
	Add Urdu notification module for policy updates	Mar–Apr 2026
	Perform final integration & testing	Apr–May 2026
	Prepare final demo & presentation materials	May 2026

Team Member	Activity	Tentative Date
	Conduct literature review & requirement analysis	Sep–Oct 2025
	Create dummy datasets simulating NADRA & bank records	Nov–Dec 2025
	Implement blockchain validation using dummy datasets	Dec 2025–Jan 2026
Abdul Moueed	Design card issuance system & prototype exploration	Jan–Feb 2026
	Assist POS demo integration with blockchain rules	Feb–Mar 2026
	Add Urdu notification module for policy updates	Mar–Apr 2026
	Perform final integration & testing	Apr–May 2026
	Prepare final demo & presentation materials	May 2026

7. Gantt Chart



8. Tools and Technologies

- **Blockchain:** Ethereum (Ganache/Remix for development); alternatives: Hyperledger, MultiChain.
- **Smart Contracts:** Solidity (Ethereum)/Chaincode (Hyperledger).
- **Backend:** Node.js or Python (Flask/Django).
- **Card Integration:** NFC reader (ACR122U) with nfcpy/pyscard libraries; alternatives: Virtual card via web/mobile app.
- **Database:** Dummy datasets (CNIC, income, household size).
- **Frontend (Optional):** HTML/CSS/Javascript for user portal.
- **Other:** Urdu text notification system.

9. References

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[6] N. Kshetri, “Blockchain’s roles in strengthening cybersecurity and protecting privacy,” Telecommunications Policy, vol. 41, no. 10, 2017.

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[9] UNDP Pakistan, “Ehsaas and BISP program evaluation report,” UNDP, 2022.

[10] "Rs141b BISP fraud exposed in audit," *The Express Tribune*, Mar. 19, 2025.